Detection and Parameter Estimation of GW150914 using a Newtonian Chirp Model

Kshitij Pandit

***Abstract*—The GW150914 signal was detected using a Newtonian chirp approximation and the open-source LIGO strain data from September 14th, 2015. The use of a Newtonian chirp allows for valid signal detection, with a least-squares fitting signal-to-noise-ratio of 83.3 from a thermal noise floor of $7.3 \pm 0.1 \times 10^{-47}\,\mathrm{Hz}^{-1}$, but appears insufficient for accurate parameter estimation due to its network $\chi^2_{\mathrm{red}} = 3.89$. The chirp mass using the Newtonian chirp is 21% larger than the official LIGO result: $36.3^{+0.8}_{-0.9}\,\mathrm{M}_\odot$, and inferred mean component masses of $41.7^{+1.3}_{-1.6}\,\mathrm{M}_\odot$.**

I. Introduction

THE detection of gravitational waves from the Livingston (L1) and Hanford (H1) LIGO sites [1] inaugurated the era of gravitational-wave astronomy. It not only verified key predictions from general relativity, but led to the first time observation of gravitational waves. The phenomenon of the merger of two black holes could not be detected without the immense precision of LIGO interferometers [2], and the depth of signal analysis to extract a clean waveform from a noise-dominated background [2], [3].

Whilst the physics behind these mergers is incredibly fascinating, it is not the topic of this paper, which assumes the general relativity model proposed is truth, and goes into depth on the signal analysis [3] to statistically prove this model - specifically its Newtonian approximation [4] works.

This report uses open source raw data from the LIGO Open Science Centre [5] and all presented analysis was done using this data.

This involves manipulating the recorded signals via whitening, band-passing, and zooming [2]; calculating the time delay between the signals [6]; applying a Newtonian chirp model [4], comparing its applicability with a χ^2 fit, and using its p-value to prove the existence of gravitational waves; and finally, graphically showing the dynamics of a cataclysmic cosmic event.

II. Theory

General relativity shows that accelerating masses produce propagating oscillations in space-time. These ripples are transverse perturbations to the spacetime metric $h_{\mu\nu}$ on Minkowski space [4]. Mathematically, these fluctuations appear as a dimensionless strain $h(t) = \Delta L/L$ (the magnitude of $h_{\mu\nu}$ [4]), which describes the change in distance between two masses.

When a gravitational wave passes through a LIGO detector, there is a tiny change in one of the arm lengths due to the expansion of space across it. Similarly, there is a small contraction in the other arm in the perpendicular arm [2]. This report describes the methods used to detect the gravitational waves from the G150914 binary black hole merger [1], which produced a strain of amplitude $\sim 10^{-21}$.

The signal from this phenomenon is modelled approximately as a Newtonian chirp - which explains the behaviour of a binary system [4]. It has the form:

$$h(t) = A(t)\cos\left[\Phi(t)\right], \tag{1}$$

where $A(t)$ is the amplitude evolution factor over time and $\Phi(t)$ is the phase evolution over time. It is given by the integral of the frequency evolution $f(t)$. The lack of Post-Newtonian terms in this report is intentional [4], and is used to exemplify the exact point Newtonian physics is no longer sufficient. The model serves as the approximation of the energy loss of a binary system, where the primary highlight is the frequency of orbit of the bodies, and how it changes in time [4]. It is noted by the model that, frequency $f(t)$ evolves via $f(t) \propto (t_{\mathrm{c}} - t)^{-3/8}$, where f is the frequency, t_{c} is the time of the collision event of the merger and t is the observer time. The entire bracketed term represents the time until the merger. The amplitude evolution $A(t)$ is of a similar form, but different exponent: $A(t) \propto (t_{\mathrm{c}} - t)^{-1/4}$.

Eq. 1 consists of $A(t)$ and the integral of $f(t)$, but the coefficients to the proportional case depend on constants such as G, c, D (see Appendix). The coefficients also depend on the *chirp mass* of the binary system - $\mathcal{M}$ [1], given by:

$$\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} \propto (f^{-11/3} \times \dot{f})^{3/5}, \tag{2}$$

where m_1, m_2 are the masses of the two binary objects before the merger and $\dot{f}$ is the rate of change of the frequency f.

The resultant graph should show three sub-phenomena within the signal, described by different stages in the frequency and amplitude evolution [1]. Initially, at relatively constant frequency and amplitude, the *inspiral* can be seen. This is the stage where both black holes orbit each other. As they approach each other and $|t_{\mathrm{c}} - t| \to 0$, the frequency and amplitude increase rapidly due to the negative exponent in the relations for $f(t)$ and $A(t)$. The next phase is the *merger*, where both black holes coalesce into one. After this, the *ringdown* is seen as the third phenomenon, where the signal rapidly drops [1]. This is because the system is stable now, with one sole black hole left.

III. Experimental Background

A. The Gravitational Wave Detector

The LIGO detectors are $\sim$ 4km long Michelson interferometers [1], shown in Fig. 1. There is also a Fabry-Pérot cavity present in each arm, which increases the effective optical path length by a factor of 300 [1]. The interferometer is in vacuum in order to effectively remove the effect of light scattering off

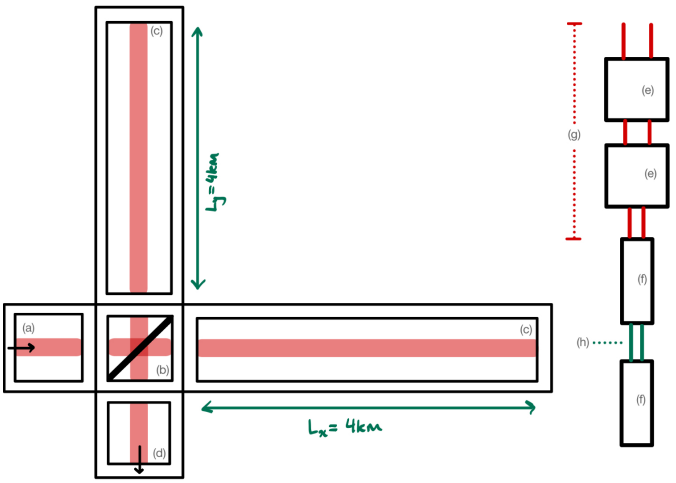


Fig. 1. The Michelson interferometer design used at both Hanford and Livingston sites [1]. The design [2], [7] consists of (a) an input laser, which passes through a (b) beam-splitter, dividing the laser into two arms of length L_{x} and L_{y} which are reflected by (c) at the end of the arm, focussing the beams into the (d) detector. On the right, the quadrupole pendulum [2] layout is shown. It consists of (e) metal masses attached by (g) steel wire to each other and (f) fused silica masses, which are both connected by (h) fused silica fibres in order to reduce the effect of thermal noise [2]

air particles. The Hanford and Livingston detectors are 'anti-aligned' [2], meaning there is approximately a phase difference of π radians between the signals produced by these detectors.

Light passes through both arms via the beam-splitter and mirror test masses. The interferometer is calibrated such that the detected signal is locked to a dark fringe [7]. When a gravitational wave interacts with the interferometer, there is a contraction in one arm whilst the other arm appears to expand, due to the perturbations in space-time [2]. This leads to a change in the phase difference, and a change in the detected signal. For the GW150914 binary, the inspiral, merger, and ringdown produce changing gravitational wave frequencies and strains, which are detected by the changing phase difference and signal strength in the interferometer [7].

B. Sources of Noise

The observed signal is masked behind layers of noise from different classical and quantum sources [2]. Most of these sources are prevalent at low frequencies, meaning a high-pass filter can be used to extract useable data, at the cost of applying a lower limit to the frequencies of detectable gravitational waves.

1) Seismic Noise: The effects of the Earth itself creates dominant sensitivity limit of the frequencies of gravitational wave detectable by these interferometers. The term seismic noise includes the effect of ocean waves hitting the coast (which can provide up to several hundred μm of displacement [2]), wind causing building shaking and human activity such as moving trucks. The optical components are mounted on an actively controlled seismic platform, which further isolates the arms from motion within the Earth by three orders of magnitude [2]. The test mass is suspended using a quadrupole pendulum setup using steel wires and three other masses, shown in Fig. 1, which absorbs the effects of ground motion by a further seven orders of magnitude [2].

2) Shot Noise: The photons inside the interferometer hit the mirror through a Poisson distribution [2]. At lower energies, there are fluctuations in the photon arrival rate [2] at which each photon arrived at the mirror, leading to noise that increases with frequency. This is linked to the Uncertainty Principle via $\Delta E \Delta t \geqslant \frac{\hbar}{2}$ [7]. By increasing the power of the laser, this would reduce shot noise, but has the issue of increasing radiation pressure noise, which is also governed by quantum effects. The Geo600 detector has investigated the effect of 'squeezing' shot noise without needing to increase the power of the interferometer, which could avoid the impact of excitations [8].

3) Radiation Pressure Noise: The Poisson distributed impacts of photons transfer momentum unto the test mass [2], causing small oscillations of the mass, leading to uncertainty in the mirrors position, demonstrating the link to the Uncertainty Principle: $\Delta p \Delta x \geqslant \frac{\hbar}{2}$, and causes low frequency noise.

4) Thermal Noise: The Brownian motion of everything in the apparatus causes displacement noise [2]. This will always exist at a small level, but can be reduced. In the quadrupole pendulum setup, the penultimate mass is attached to the test mass via fused silica fibres, which have a high mechanical quality factor and reduce the impact of thermal noise [2]. The test mass is a mirror made from fused silica for the same reason [2].

5) Newtonian Noise: Gravitational forces from local changes in the Earth's crust also causes noise. When a seismic wave passes through the ground, it causes a minute change in density in the ground near the test mass, which affects the gravitational pull on the mirror. This leads to a small, sub 20 Hz displacement noise, which is masked by seismic effects [2]. Theoretically, if an interferometer were to be seismically isolated, Newtonian noise would be the floor [2], and could only be reduced by removing large gravitational effects. The proposed Laser Interferometer Space Antenna (LISA) [9] circumvents the effects of both seismic and Newtonian noise and could operate in the millihertz regime (0.1 mHz - 1 Hz).

IV. Data Analysis Methodology

Information about how the open source data [5] was accessed can be found in the acknowledgements. The open source files contained 4096 frequency points, at a sample rate of 4096 Hz, meaning each point was a whole number frequency value; from this, the error in frequency is clearly ± 0.5 Hz. From the sample rate, the temporal resolution error was recorded as ± 0.0001s.

The data from both sites are vertically aligned to have strain centred at 0. This is done in order to more easily compare the following aspects: the signal shapes observed at both detectors, if there is any (significant) time delay, Δt and if there is any significant amplitude variation. The absolute magnitude of the strain is not relevant, but the relative magnitudes within a detector's signal is. This is because the Newtonian quadrupole has an increasing amplitude and frequency [4] as the two modelled bodies approach collision.

A. Whitening

The *signal-to-noise-ratio* (SNR), ρ is useful in quantifying the uncertainty in chirp mass $\mathcal{M}$ via $\frac{\sigma_{\mathcal{M}}}{\mathcal{M}} \approx \frac{1}{\rho}$ [4], [10], and hence a larger ρ yields a smaller $\sigma_{\mathcal{M}}$ by inverse proportionality. A fast Fourier transform [3] of the time domain can be

taken, from which the apparent 'loudness' of certain frequencies can be compared with others. Each frequency's power spectrum can be normalised, meaning that every frequency has an equal impact on the finalised signal. An inverse transform is used to return to the time domain and the whitened strain $\tilde{h}(t)$.

B. Defining Noise Boundaries

The effects of seismic, radiation pressure and Newtonian noise are all prevalent at lower frequencies. These frequencies can be removed to give a more relevant frequency sample to detect the chirp in.

Welch's method [3] can be used to generate a smooth frequency plot. From this plot, a simple extraction algorithm can be used to quantify the frequency ranges which correspond to noise, shown in Fig. 2. This resulted in a value of 36 Hz.

The maximum frequency of the frequency is dependent [4] on the frequency of the binary orbit at the end of the inspiral (known as the innermost stable circular orbit - ISCO) and the frequency at the start of the ringdown [1]. The ISCO frequency and the 'bell' frequency at the start of the ringdown were measured as 67.6 ± 0.5 Hz and 184.6 ± 0.5 Hz respectively. A safety margin was applied, extending the maximum frequency to 300 Hz.

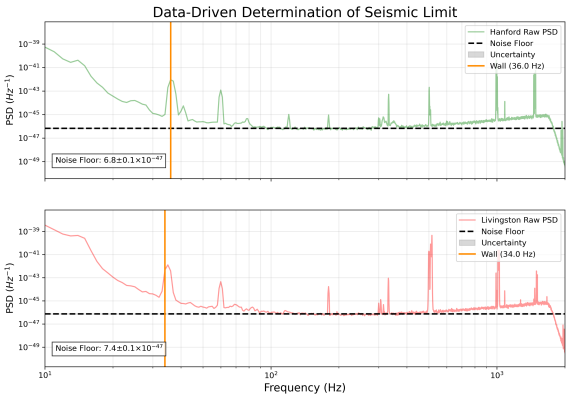


Fig. 2. Two Welch periodograms of frequencies and their power spectral densities [3] for both sites. There are two regions of increased frequency at the ends of both plots, with the left-hand region completely dominated by seismic noise. There are two artifacts in this region, one being the large 'mountain' beginning at 22 Hz, which corresponds largely to oceanic seismic noise, pendulum resonances [2], and other low-frequency sources. There is another artifact at 36 Hz and 34 Hz for H1 and L1 respectively corresponding to a 'foothill.' All frequencies below the the mean foothill value are to be removed, treating 35 Hz as a *safety margin*. There is also a smaller increase in frequency on the right-hand side of the graph due to quantum shot noise [2]. Its effects only become significant at ≈ 350 Hz. The drop leading up to 2048 Hz is likely due to an aggressive low-pass anti-aliasing filter [5], [6] applied before data sampling. At frequencies above 500 Hz, spiking violin modes of the quadrupole pendulum system can be seen. Similarly, the US power grid causes sharp peaks at 60 Hz and its harmonics [1].

By sampling the quiet, thermal noise driven region [2] chosen to be $f \in [35, 300]$ Hz, a thermal noise floor can be defined using the median value and median absolute deviation [11] to avoid distortion from 60 Hz power mains and harmonics [1]. The noise floor was estimated as $S_n = 7.3 \pm 0.1 \times 10^{-47} \mathrm{Hz}^{-1}$ using an inverse variance weighted mean [12] of the floor of both sites.

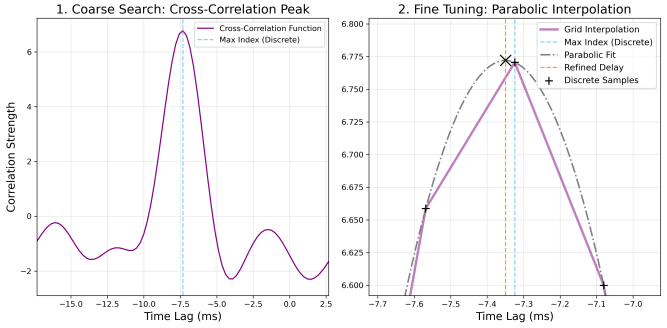


Fig. 3. The left-hand figure shows each time lag plotted against it's respective correlation. This was achieved by sliding [6] the Hanford signal over Livingston's for a discrete lag value corresponding to the temporal resolution error. At each point, the correlation of the models was calculated [13] and plotted. The discrete maximum delay $\tau_{\mathrm{grid}} = 7.32 \pm 0.01$ ms is the maximum discrete lag index, and is plotted with a blue dashed line. The right-hand figure shows the parabolic interpolation method used to transcribe discrete data into an estimate for the continuous true value, with the new, refined prediction marked with the orange dashed line.

C. Bandpassing and Zoom-In

A Butterworth [6] filter was applied at order 4 to isolate the calculated frequency range. Higher orders were forgone in order to reduce the impact of the filter creating unwanted signal shapes [6].

From the resulting data, an amplitude spike in the band-passed strain was detected in the range of 16.2187 - 16.5187 s with a temporal resolution of $\pm 0.0001s$. A graphical zoom can now be justified to within this region, which revealed two signals that resembled an inspiral, merger, and ringdown [2].

D. Temporal Alignment

Both LIGO sites are over 3000 km from each other. The gravitational waves were detected first at Livingston, and after a delay, at Hanford [1]. In order to accurately determine the time delay, the cross-correlation method [6] was used along with interpolative methods to bypass the discrete nature of the data. This method accurately worked out the time delay by mathematically sliding the strain of one signal over the other [13], generating the time lag plot seen in Fig. 3. The lag value with the greatest correlation strength is the maximum discrete value $\tau_{\mathrm{grid}} = 7.32 \pm 0.01$ ms, limited by the sampling rate. The two adjacent points are used to generate a parabola, from which interpolation [14] is used to determine a systemic bias correction $\delta t = 0.03 \pm 0.01$ ms. Applying the correction $\tau_{\mathrm{true}} \approx \tau_{\mathrm{grid}} + \delta\tau$, The new maximum $\tau_{\mathrm{true}} = 7.34 \pm 0.01$ ms is used as the time delay between the two detectors. This removes the temporal resolution error; the new error now is only limited to the observatory synchronisation at a magnitude of $\sim 10\mu$s [1], [2].

E. Curve-Fitting

After using the calculated time delay to align the signals from both sites, Eq. 1 was applied to the data. From this, The chirp mass $\mathcal{M}$ and merger time t_{c} can be estimated. Due to the Newtonian approximation [4], frequencies cannot be calculated closer than 0.07 s to the merger; it is here where Post-Newtonian terms [4] are needed to fit at times closer to t_{c}. Numerical relativity is needed to fit the actual merger event at $t = t_{\mathrm{c}}$ [1], [4].

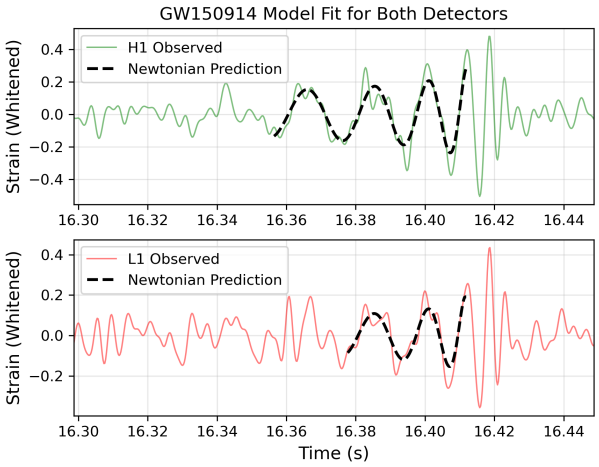


Fig. 4. The final, overlapped signal for both sites. Clearly, an oscillation is visible for times $t < t_c$ (correlating to the inspiral [1]). Close to the event, the effect of the frequency and amplitude evolutions are seen as a sharp increase in both quantities. At the maximum amplitude, the merger [1] event $t = t_c$ occurs. Beyond this, there is a sharp drop to null signal, signifying the ringdown stage [1]. Eq. 1 is fitted to a small sample of the signal. The entire chirp is not fully fitted because the Newtonian approximation fails at closer times to the event [4], due to the lack of relativistic treatment. The real process is more violent and hence the evolution of frequency is faster - causing a phase mismatch between signal and model.

V. Results and Discussion

1) Time Misalignment and Delay: The persistence of a non-zero event time delay $\Delta t = 1.5 \pm 0.1\,\text{ms}$ after temporal alignment is a sign that the Newtonian approximation fits the inspiral phase slightly differently for each detector, likely due to unmodelled merger dynamics [4] and the assumption that the Hanford detector orientation is a perfect π radians phase inversion, which may not be exact. The times are consistent within 2.2σ The estimate for the true time delay $\tau_{\text{true}} = 7.34 \pm 0.01$ ms is an improvement to the discrete maximum τ_{grid} but is not guaranteed to be the true value: it is limited by the assumption that the discontinuous peak of cross-correlation is parabolic in nature.

2) Goodness of Fit and Detection Statistics: To quantify the fidelity of the Newtonian model, the reduced χ^2 value was calculated for the inspiral of both detector fits, resulting in individual values of reduced $\chi^2_{\text{H1}} = 3.72$ and reduced $\chi^2_{\text{L1}} = 4.06$. The joint network had a resultant value of $\chi^2_{\text{red}} = 3.89$. These values are not near unity, and hence confirm the Newtonian model requires Post-Newtonian terms for a numerically accurate model.

Whilst being numerically inaccurate, the model confidently demonstrates the existence of the chirp signal. We define the null hypothesis H_0 that the detector only outputs thermal noise post-signal-processing; least squares model fitting yields the test statistic: the SNR ρ, with $\rho_{\text{H1}} = 67.6$ and $\rho_{\text{L1}} = 48.6$, summing in quadrature to the network SNR, $\rho_{\text{net}} = 83.3$. The associated p-value for this test statistic is on the magnitude of 10^{-1500}. This provides an overwhelming amount of evidence to reject the null hypothesis, and it far exceeds the standard threshold of 5σ required for discovery. This is now sufficient to verify the existence of gravitational waves and constitutes as the first physical detection of a black hole merger [1] in the form of GW150914.

TABLE I
Results from Data Analysis

Parameter	H1	L1	Combined/ Difference
Noise Floor (S_n) 10^{-47} /Hz^{-1}	6.9	7.7	7.3
Merger Time (t_c) /s	16.4160	16.4145	Δt =0.0015
Time Delay (τ) /ms	—	—	7.34
Chirp Mass ($\mathcal{M}_{\text{source}}$) /M$_\odot$	35.6	37.4	36.3
Reduced χ^2	3.72	4.06	3.89
SNR (ρ)	67.6	48.6	83.3

3) Chirp Mass and Parent Mass: The chirp mass of the system can be measured by curve fitting the full form of Eq. 1. The Hanford data yielded a measurement of $\mathcal{M} = 38.9\,\text{M}_\odot$, whilst $\mathcal{M} = 40.8\,\text{M}_\odot$ was measured by Livingston's model. Using an SNR-weighted mean [12] to achieve $\mathcal{M}_{\text{detector}} = 39.5 \pm 0.5\,\text{M}_\odot$ and applying a redshift factor $z = 0.09^{+0.03}_{-0.04}$ [15], the source mass chirp mass can be measured by the relation: $\mathcal{M}_{\text{detector}} = (1+z)\mathcal{M}_{\text{source}}$ [15]. Note the official result for z stated here cites 90% credible intervals instead of 1σ [1], [15]. This results in a final, source chirp mass: $\mathcal{M}_{\text{source}} = 36.3^{+0.8}_{-0.9}\,\text{M}_\odot$. The official LIGO value [1], [15] claims $\mathcal{M}_{\text{source}} = 30\,\text{M}_\odot$. The offset is likely due to the imperfect fit of the Newtonian model ($\chi^2_{\text{red}} = 3.89$); this is probably due to the Newtonian model perceiving the relativistic plunge into the merger as a faster inspiral. The chirp mass alone can be used to estimate the parent black hole masses (assuming an equal mass binary system) using Eq. 2. The masses were estimated to be $m_{1,2} \approx 41.7^{+1.3}_{-1.6}\,\text{M}_\odot$, which are significantly larger than the literature values of $36\,\text{M}_\odot$ and $29\,\text{M}_\odot$. The error ($\sim 1.6\,\text{M}_\odot$) is much smaller than the discrepancy from these values $\sim 8\,\text{M}_\odot$, and varies by $\sim 5\sigma$, providing evidence the error is a direct propagation of the systematic Newtonian chirp bias [4].

VI. Conclusion

The discovery of the GW150914 signal not only acts as the first significant observation of gravitational waves, but also as the first detection of a binary black hole merger. The predicted masses, $m_{1,2} = 41.7\,\text{M}_\odot$ of the parent black holes provide evidence of large stellar mass black holes. The network signal-to-noise-ratio $\rho_{\text{net}} = 83.3$ was large enough to reject the null hypothesis of only thermal noise with a p-value of false alarm far exceeding the discovery threshold of 5σ of ($p << 10^{-7}$). Post-Newtonian relativistic corrections were forgone by the model, meaning the Newtonian chirp was insufficient for valid parameter estimation, with the network $\chi^2_{\text{red}} = 3.89$. This is clear in the presence of non-zero time deltas within the two sites reported time of merger, and the model's reported value for chirp mass $\mathcal{M} = 36.3$ which is an overestimation of 21%. This means, whilst the Newtonian model is sufficient enough to discover gravitational waves, it is not sufficient for accurate parameter estimation.

APPENDIX
THE FULL NEWTONIAN CHIRP

The full Newtonian Chirp relation is given by adding extra terms to the amplitude evolution. This gives the following equation:

$$h(t) = -\left(\frac{G\mathcal{M}}{c^2 D}\right)\left(\frac{c^3 A(t)}{(5G\mathcal{M})^{-1/4}}\right)\cos\left[\Phi(t)\right], \qquad (3)$$

where G is the gravitational constant, $\mathcal{M}$ is the chirp mass, c is the speed of light and D is the distance to source.

The phase relation $\Phi(t)$ is given by the integral of the angular frequency $\omega(t) = 2\pi f(t)$ [4]:

$$\Phi(t) = \int 2\pi f(t) dt \qquad (4)$$

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